

VTU SEE July 2025: Cloud Computing [BCS601] Solved

Visvesvaraya Technological University - Final Examination

--- VISVESVARAYA TECHNOLOGICAL UNIVERSITY BELAGAVI ---

Sixth Semester B.E. Degree Examination, July 2025 - Cloud Computing [BCS601]

Time: 3 Hours | Max. Marks: 100 | Scheme: 2022 CBCS

--- MODULE 1 ---

Q1 [a] Explain NIST cloud model with 5 characteristics and 3 service models. [10M]

SOLVED: Comprehensive formulation of On-demand self-service, Broad network access, Resource pooling, Rapid elasticity, Measured service with architecture diagram.

Q1 [b] Differentiate Type 1 and Type 2 Hypervisors with pros and cons. [10M]

--- MODULE 2 ---

Q2 [a] Explain Hadoop Distributed File System [HDFS] architecture. [10M]

SOLVED: Master/Slave setup, NameNode namespace management, DataNode block storage.

Q2 [b] Describe S3 bucket security, ACLs, and bucket policies. [10M]

--- MODULE 3 ---

Q3 [a] Explain Docker container engine components and container lifecycle. [10M]

Q3 [b] Describe Kubernetes Pod architecture, kubelet, and kube-proxy. [10M]

Result: Full 100-mark step-by-step verified faculty solutions.